

reaches 0.267 m s^{-1} in the opposite direction.

- 3 (a) (i) The direction of the resultant force acting on the object must be perpendicular to the direction of its velocity and towards the centre of the circular path.
- (ii) As the direction of the motion of the object is constantly changing, its velocity is constantly changing. Since acceleration is the rate of change of velocity, it has an acceleration directed towards the centre of the circular path.

(b) (i) $F_G = F_C$

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$r = \frac{GM}{v^2}$$

$$r = \frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{2500^2} = 6.4 \times 10^7 \text{ m} = 6.4 \times 10^4 \text{ km}$$

- (ii) 1. The potential energy of the satellite is 0 at a large distance from Earth. As the radius of the orbit is reduced, the potential energy of the satellite becomes more negative, hence decreases.
2. The kinetic energy of the satellite increases, as the radius is reduced from the expression $\frac{1}{2}mv^2 = \frac{GMm}{2r}$.